A Generalized Metric for Quantifying Skill vs. Luck Across All Competitions

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1. Introduction

The fundamental appeal of competitive sports lies in the "Uncertainty of Outcome." A competition determined entirely by skill would be predetermined and devoid of dramatic tension, while one determined entirely by luck would fail to sustain spectator interest, as the outcomes would be perceived as meaningless random events rather than sporting achievements. Consequently, quantifying the precise balance between these two forces is not merely an academic exercise; it is a central problem in sports analytics with profound implications for league design, revenue distribution, and the evaluation of competitive integrity. Understanding where a specific sport lies on the "Skill-Luck Continuum" allows governing bodies to optimize league formats to ensure that the best team wins while maintaining enough variance to keep fans engaged.

To measure this balance, researchers have proposed various quantitative methodologies. Early efforts, such as the Noll-Scully ratio [1], attempted to compare the observed variance of win percentages to the theoretical variance of a coin-flip league. Beyond this classical competitive-balance perspective, Mauboussin's conceptual work on the "skill–luck continuum" emphasized that observed performance variance can be decomposed into stable ability and random noise components across different sports [2]. More recently, Aoki et al. [3] introduced a distance-based coefficient that compares observed league standings with those of an idealized perfectly skill-based competition, while Lopez et al. [4] developed Bayesian state-space models using betting-market data to disentangle persistent team strength from game-to-game randomness across multiple North American leagues. In parallel, Csurilla et al. [5] systematically compared several luck metrics and competitive-balance indicators in basketball, defining luck as the discrepancy between expected and realized outcomes and positioning different sports (and formats) on a common scale. Building on this, Getty et al. [6] introduced the R-metric, an approach rooted in the principle of variance decomposition**.** By analytically isolating the variance attributable to skill from random noise, this framework quantified the contribution of skill in fantasy sports and established a quantitative baseline for comparing the relative roles of skill and chance across diverse activities. Due to its intuitive framework and mathematical robustness, the R-metric has since been widely recognized as a generalized standard for evaluating competitive balance across a broad spectrum of games and contests.

However, the traditional R-metric suffers from a critical limitation: it relies on the assumption of temporal persistence. Getty's methodology typically splits a season chronologically, such as comparing the first half against the second half, to measure performance consistency. This approach implicitly assumes a balanced schedule where opponent difficulty is uniformly distributed over time. This assumption fails in competitions with unbalanced schedules. In formats such as major tournaments (e.g., FIFA World Cup, UEFA Champions League), split-leagues (e.g., Scottish Premiership), and individual sports (e.g., Tennis, Golf), participants face a unique and varying set of opponents. Previous studies in sports economics have shown that unbalanced schedules can systematically distort both competitive-balance measures and reported league tables, because teams do not face opponents of comparable strength with equal frequency [7–9]. In these scenarios, the temporal splitting method cannot distinguish between a fluctuation in a team's intrinsic performance and a variation in opponent difficulty. Consequently, schedule luck confounds the measurement, rendering the traditional R-metric inapplicable to a vast array of modern competitive formats.

To overcome these limitations and provide a unified standard for skill quantification, this study presents the following contributions:

1. **Proposal of Generalized R-Metric:** We propose a Generalized R-metric that replaces the traditional temporal splitting with Stratified Splitting based on the ELO rating system [10]. By estimating opponent difficulty using dynamic rating systems and partitioning matches based on difficulty rather than time, we effectively control for schedule bias in unbalanced formats.
2. **Dual-Strategy Framework:** We introduce a robust validation framework combining Getty's R with Artificial Round-Robin Benchmarking. By cross-validating the results derived from our generalized metric against structurally balanced "virtual leagues" constructed from historical data, we demonstrate the statistical validity of our findings across both team and individual sports.
3. **Comparative Analysis with Open Benchmark Across Diverse Domains:** We conduct an extensive comparative analysis across a wide range of competitive environments, encompassing European and US sports leagues, individual and team sports, and both league and tournament formats. To the best of our knowledge, this study provides one of the first openly available benchmark datasets[1] of this scope for unified skill–luck analysis, together with a fully reproducible evaluation pipeline. This establishes a unified "Skill–Luck Spectrum" on a single standardized scale and enables replication, transparent comparison, and future extensions across sports.

The remainder of this paper first presents the methodological foundations of the proposed dual strategies in Section 2, followed by extensive experimental validation across synthetic data, team sports, and individual sports in Section 3. Section 4 then discusses the practical and strategic implications of the results for competition design and sports governance.

2. Methodology

This section establishes a mathematical framework to quantify skill across diverse competitive formats, including tournaments and individual sports. We first generalize the theoretical foundation

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[1] https://github.com/leemingo/skill-vs-luck

of the R-metric to accommodate continuous performance indicators. We then propose two complementary approaches to implement this metric: first, we utilize ELO-based Stratified Splitting for robust analysis of full datasets, and second, we construct Artificial Round-Robin specifically to address scenarios characterized by data sparsity or where reliable strength estimation (ELO) is infeasible.

2.1. Theoretical Foundation of the R-Metric: Variance Decomposition

The core of the R-metric relies on decomposing the variance of competitive outcomes into components attributable to skill and luck. As established by Getty et al., we define the performance of an entity i in two statistically equivalent subsets of matches as P_i and Q_i. These values are transformed into a rotated coordinate system (S, T) defined as:

$$S_i = \frac{P_i + Q_i}{\sqrt{2}},\ T_i = \frac{Q_i + P_i}{\sqrt{2}} \tag{1}$$

Here S_i represents the skill component (consistent performance across subsets), while T_i represents the luck component (inconsistent variation). The R-value is derived from the ratio of the variances of these components across the population:

$$R = 1 - \frac{Var(T)}{Var(S)} \tag{2}$$

2.2. Strategy I: Generalized R via Stratified Splitting

This strategy is designed for the robust analysis of full datasets where an objective measure of opponent strength is available. It replaces temporal splitting with a difficulty-based stratification process.

2.2.1. Opponent Strength Estimation via ELO

We first estimate the relative strength of each team using the Elo rating system [2]. After each match between team i and team j, Elo ratings are updated as:

$$\theta^*_i = \theta_i + K(O_i - E_i),\ \theta^*_j = \theta_j + K(O_j - E_j), \tag{3}$$

,where θ_i is team i's pre-match rating, K is the update factor, $O_i \in \{1,\ 0.5,\ 0\}$ is the observed outcome, and $E_i = \frac{1}{1+10^{(\theta_j - \theta_i)/400}}$ is the expected win probability. This process is iterated chronologically through all matches to generate dynamic strength estimates for each team across the season. In this study, we set the update factor to $K = 20$.

2.2.2. Stratified Splitting by Opponent Strength

Instead of splitting seasons into temporal halves, we partition each team's matches into two folds such that the average opponent Elo in each fold is statistically balanced. Formally, let a team's match set be $M_i = \{m_1, m_2, \cdots,\ m_n\}$, with opponent ratings $\{r_1, r_2, \cdots,\ r_n\}$. We seek two disjoint subsets $A_i,\ B_i\ \in M_i$ with:

$$\left| \frac{1}{|A_i|} \sum_{m \in A_i} r_m - \frac{1}{|B_i|} \sum_{m \in B_i} r_m \right| < \epsilon, \tag{4}$$

where ϵ is a small tolerance ensuring fold-level difficulty is approximately equal, and it can be used to control for schedule imbalance.

2.2.3. Aggregation

For each split, we compute fold-level performance (x_i ,y_i), apply the standard R rotation, and repeat the stratified splitting K times, reporting the generalized metric as the split-average.

2.3. Strategy II: Artificial Round-Robin Benchmarking

While Strategy I offers a generalized solution for dense datasets, it faces limitations in environments characterized by data sparsity or where the tuning of dynamic rating systems is methodologically infeasible due to insufficient connectivity between participants. Strategy II addresses these constraints by constructing a structurally balanced subset of data.

2.3.1. Construction of Virtual Leagues via Temporal Aggregation

To overcome the data sparsity inherent in unbalanced or short-season formats, we construct "Artificial Round-Robin" datasets by aggregating multi-year data. For example, in individual sports such as tennis or golf where head-to-head records for lower-ranked players are sparse, we filter for the top N participants over a multi-year window and retain only the matchups where every pair has faced each other at least twice. This process creates a dense, structurally idealized sub-graph of the competition, effectively simulating a balanced league format from historical data. The respective time-window is chosen so that team fluctuation is minimal and that age-related performance variations can be ignored.

2.3.2. Calculation of R-Metric on Virtual Leagues

Since the Artificial Round-Robin dataset effectively neutralizes structural imbalance to mimic a balanced schedule, the traditional R-metric framework can be applied directly without ELO-based stratification. We simply partition the aggregated matchups into statistically equivalent subsets (e.g., via random assignment) and compute the performance metrics. This approach provides a model-independent quantification of skill for the selected high-density sub-population, serving as a robust alternative when dynamic strength estimation is challenging.

2.4. Data Description

To validate our framework, we utilized a comprehensive dataset spanning diverse competitive domains, including major European football leagues, Handball-Bundesliga (HBL), North American sports (NBA, MLB, NFL), and individual sports (Tennis, Golf) over the last decade. Additionally, specific competitions were selected to test the framework under structural imbalance: the UEFA Champions League (UCL) serves as a validation case for knockout tournament formats, while K League 1 is included to assess the metric's efficacy in split-system (i.e., unbalanced) leagues. Detailed specifications of the dataset, including competition formats, participant numbers, and match volumes, are provided in Appendix A.

3. Experimental Results

In this section, we present the results of our comprehensive validation process. We first verify the mathematical correctness and robustness of the Generalized R-metric using synthetic datasets where the ground truth is known (Section 3.1). Subsequently, we confirm its compatibility with existing methods in balanced environments (Section 3.2) and extend the validation to the complex domain of individual sports using our dual-strategy framework (Section 3.3).

3.1. Theoretical Validation via Synthetic Simulation

Before applying our framework to empirical data, we rigorously tested the mathematical integrity of the Generalized R-metric using Monte Carlo simulations. By generating synthetic agents with latent intrinsic strength parameters used solely to define ground-truth win probabilities, we isolated and evaluated the metric's response to structural biases, specifically schedule imbalance. To verify whether our proposed method can recover the true skill parameter $(R_{true} \approx 1.0)$ even under extreme schedule disparities, we simulated a league of 20 agents with fixed intrinsic pairwise win probabilities ranging from 0.05 to 0.95, where this range represents their relative competitive strength in pairwise matchups (e.g., if Team A is associated with 0.05 and Team B with 0.10, Team A has a lower probability of winning against Team B). We systematically manipulated the Degree of Schedule Imbalance (δ) from 0.0, representing a random assignment or balanced schedule, to 1.0, representing a sorted schedule where agents face the weakest opponents in the first half and the strongest in the second. Full construction details are provided in Appendix B.

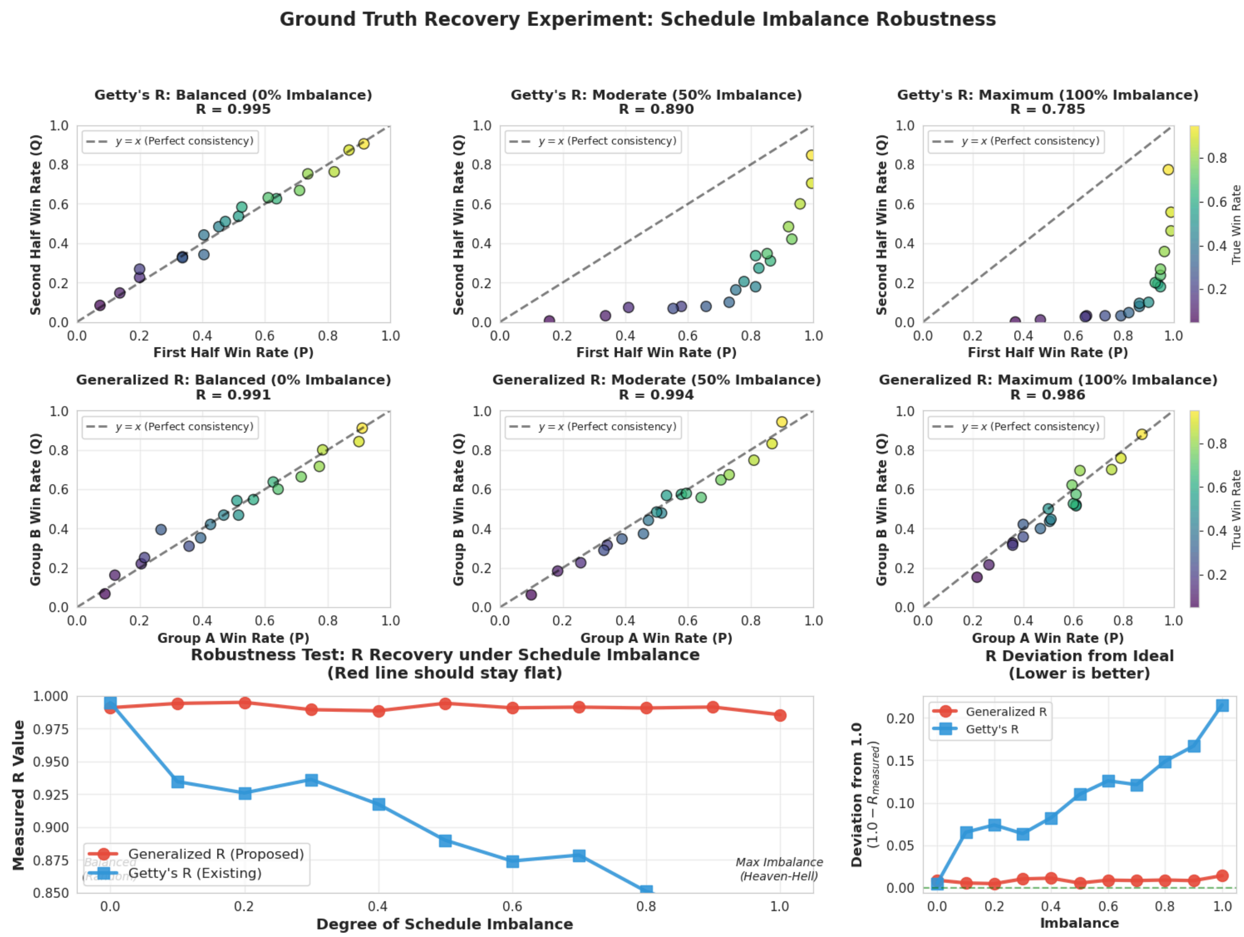

Figure 1. Robustness of Skill Estimation against Schedule Imbalance. Top and middle rows contrast the deviation of Getty's R with the consistency of Generalized R under increasing schedule imbalance. The bottom row confirms the superior stability of the proposed metric (red) compared to the baseline (blue).

The comparative analysis reveals a stark contrast between the two methods. As shown in Figure 1, the traditional Getty's R degrades significantly as imbalance increases; at maximum imbalance (δ =1.0), data points deviate from the identity line ($y = x$), causing the R-value to drop from 0.995 to 0.785. This failure stems from the temporal splitting method misinterpreting schedule-induced performance dips as skill inconsistency.

In contrast, the middle and bottom row demonstrates that the Generalized R-metric effectively neutralizes schedule bias. Regardless of imbalance levels, data points remain tightly clustered along the identity line with R-values consistently maintained above 0.98. The robustness test confirms this stability, showing a flat trajectory for Generalized R compared to the steep decline of the traditional metric. This conclusively demonstrates that the proposed opponent-strength–based stratified splitting successfully decouples skill from schedule difficulty, ensuring accurate quantification even in highly unbalanced environments.

3.2. Empirical Baseline Comparison in Balanced Leagues

Having validated the theoretical robustness of the Generalized R-metric via simulation, we now apply the framework to real-world data across 16 major sports leagues to assess its empirical consistency and adaptability. For each competitive domain, we utilized match data spanning approximately the most recent 10-year horizon, with R-values computed separately for each season to reduce the influence of team fluctuation and aging effects. This analysis serves a dual purpose: first, to demonstrate backward compatibility, ensuring that our method produces results consistent with the traditional Getty's R in balanced environments; and second, to highlight structural diagnosis, revealing necessary divergences in unbalanced formats where traditional methods fail.

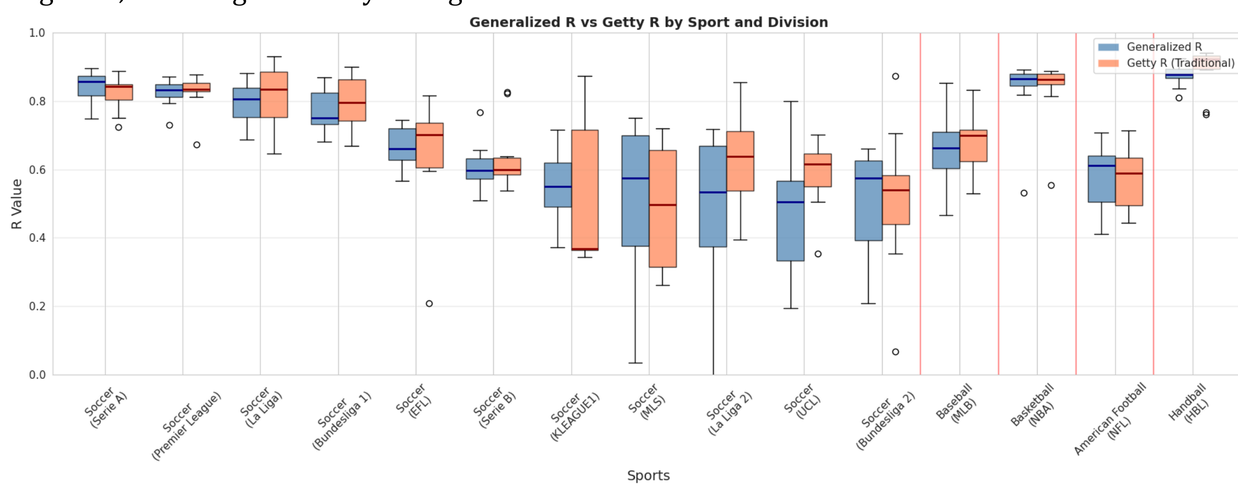

Figure 2. Empirical Comparison of Generalized R vs. Getty's R across Diverse Domains. Boxplots demonstrate strong alignment in balanced leagues, confirming backward compatibility. In unbalanced formats (K League & UEFA Champions League (UCL)), Generalized R corrects for schedule bias, diverging from the traditional metric.

Figure 2 presents a side-by-side comparison of the Generalized R (blue) and Getty's R (orange) across various divisions. In major European football leagues characterized by standard double round-robin schedules—such as Serie A, the Premier League, La Liga, and Bundesliga—the two metrics exhibit remarkable alignment. This strong concordance confirms that our stratified splitting method successfully mimics the natural temporal balance of these leagues, demonstrating that the Generalized R is fully compatible with established methods in standard environments. Interestingly, within the domain of soccer, we observe a positive correlation between league hierarchy and skill dependence. Top-tier divisions (e.g., Premier League, La Liga) consistently display higher R-values compared to their respective second divisions (e.g., EFL Championship, La Liga 2). This suggests that at elite levels, teams possess greater consistency and tactical maturity to mitigate randomness, whereas lower divisions are characterized by higher parity and volatility.

Crucially, however, significant discrepancies emerge in leagues with structural imbalances. A clear divergence is observed in K-League 1, where Getty's R overestimates variance due to the mid-season split system. Similarly, in the UEFA Champions League (UCL), Generalized R provides a necessary correction for the "randomness of the draw." Furthermore, the absolute R-value for the UCL is noticeably lower than that of domestic leagues. This quantitatively confirms that knockout-style tournaments are inherently more susceptible to luck—a single unfavorable match or draw can eliminate a superior team, unlike in long-form leagues where skill converges over time.

Finally, extending the analysis across sports reveals a distinct hierarchy on the Skill-Luck Spectrum. High-scoring sports like Basketball (NBA) and Handball (Handball-Bundesliga, HBL) exhibit the highest R-values, indicating a dominance of skill driven by large sample sizes of scoring events. In contrast, Soccer and Baseball (MLB) occupy a middle ground, where the scarcity of scoring events amplifies the relative weight of random variance. This unified comparison validates the Generalized R not only as a correction tool for specific formats but as a standardized metric capable of characterizing the fundamental nature of diverse competitive domains.

3.3. Sensitivity of the R-Metric to Competition Density

While Generalized R effectively neutralizes schedule bias, the density of competition—specifically the number of matches and participants—remains a critical factor influencing the R-metric. To empirically investigate this sensitivity, we performed a Top-Tier Slicing experiment using historical data from the four major European football leagues (Premier League, La Liga, Bundesliga, and Serie A). For the Top-*N* team conditions, we systematically vary the number of retained teams N, filter

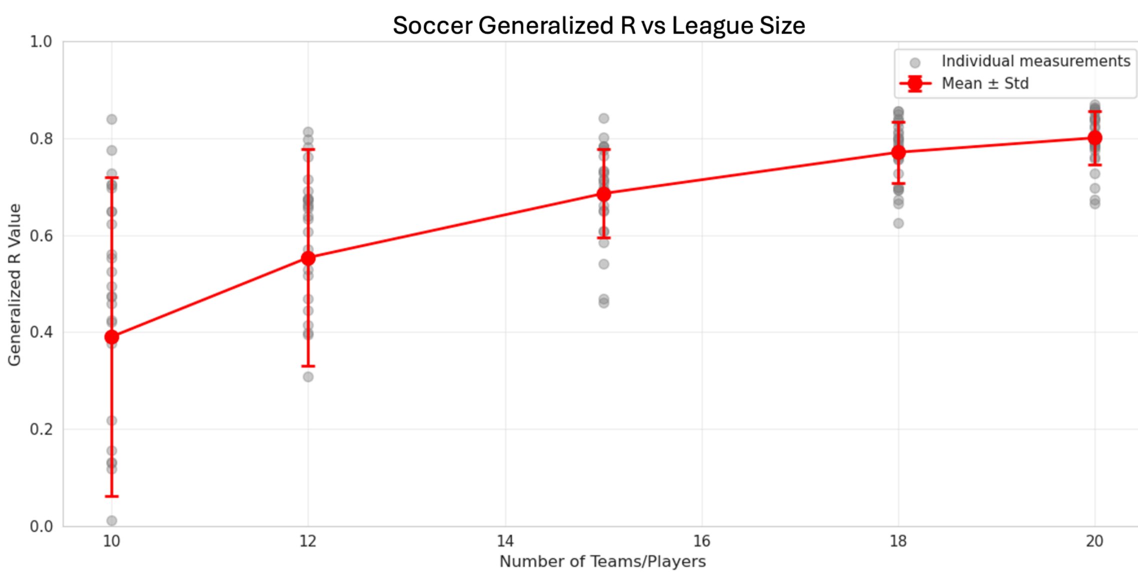

Figure 3. Analysis of Generalized R-values across four major European leagues as the sample size is

restricted to the top tier ($N = 20$ to 10). The result demonstrates that insufficient data volume leads to an underestimation of skill.

each season to retain only the top-ranked N teams, and recompute the Generalized R-value using only matches played among this reduced set, thereby constructing restricted round-robin subsets of varying density.

As illustrated in Figure 3, the results reveal a strong dependency between the sample size and the measured R-value. When the full roster of 20 teams is included, the R-value stabilizes at a high level, reflecting the true dominance of skill across the league. However, as the sample size is artificially reduced to the elite tier (e.g., Top 10), the R-value exhibits a sharp systematic decline. This trend confirms that calculating R based on a small, homogeneous subset of participants fails to capture the broader skill variance, leading to an underestimation of the sport's true skill component. This empirical finding provides conclusive evidence that a sufficient volume of data is a prerequisite for robust skill quantification. In scenarios characterized by data sparsity, such as individual sports where top competitors rarely face each other, the signal of skill is easily obscured by noise if the sample size is too small. Consequently, this validates the necessity of data aggregation strategies, such as the Artificial Round-Robin, to ensure that the dataset is sufficiently large to recover the latent skill parameter.

3.4. Validation in Individual Sports using Artificial Round-Robin

Skill quantification in individual sports is fundamentally challenging due to the absence of fixed league schedules and the resulting sparsity of direct head-to-head matchups among elite players. Unlike team sports, top competitors in tennis and golf do not systematically face one another within a single season. To overcome this structural limitation, we apply the Artificial Round-Robin strategy to historical Tennis and Golf data. By constructing fully connected, structurally balanced virtual leagues, this approach enables the direct use of the classical Getty's R-metric as a model-independent benchmark. This allows us to empirically verify the latent skill dominance of individual sports once data sparsity is properly resolved. Domain-specific constructions for Tennis (head-to-head) and Golf (stroke play) are described below.

3.4.1. Experimental Results in Tennis Data

To evaluate how interaction density affects skill estimation in individual sports, we applied the Artificial Round-Robin framework to ATP Tour match data from 2014 to 2023 and computed the classical R-metric for the Top 40 players under different temporal aggregation schemes. With 40 players, a fully connected round-robin would consist of 39 match-days with 20 matches each (780 total matches), ensuring that every player faces every other player exactly once. Each point in Figure 4 represents the R-value computed from a single time block (e.g., 2014–2015, 2015–2016), where match data within each block are pooled to construct an Artificial Round-Robin before R calculation. As shown in Figure 4, single-season estimates (green circles) exhibit strong volatility and systematically low R-values, ranging from near zero to approximately 0.65, reflecting the extreme sparsity of head-to-head matchups within a single year.

As data are aggregated over 2-year windows (yellow squares) and 3-year windows (purple triangles), both the number of matches and the R-values increase monotonically, with R stabilizing in the range of approximately 0.65–0.78 once the sample size exceeds roughly 500–900 matches. The full 10-year aggregation (red star) yields the highest R-value, representing the asymptotic skill-dominant regime under maximum interaction density. However, such long-horizon pooling may introduce slow-scale confounders such as aging and performance drift. Consequently, 2–3 year pooling emerges as the most reliable practical regime, providing sufficient statistical stabilization while preserving quasi-stationary player skill.

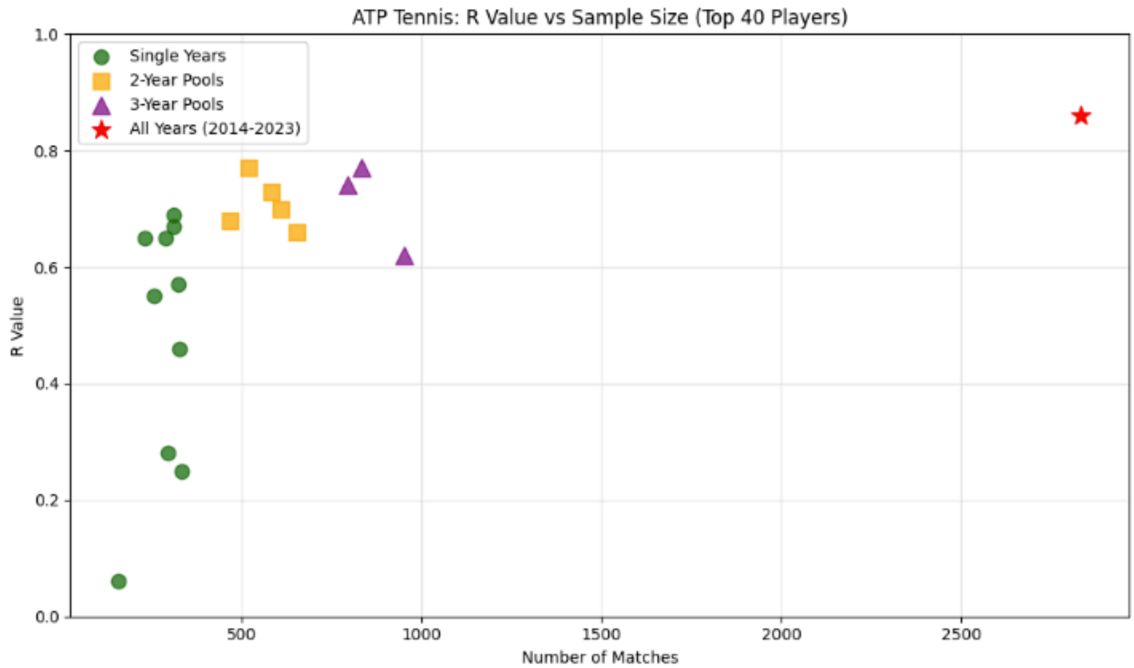

Figure 4. R-value as a function of sample size for ATP Top 40 players under different temporal aggregation schemes.

3.4.2. Experimental Results in Golf Data

Unlike tennis, professional golf is conducted in a stroke-play format without direct head-to-head elimination. To construct an Artificial Round-Robin benchmark in this setting, we generated artificial head-to-head matchups by pairing players within each tournament based on relative finishing positions. We systematically varied the number of artificial pairing rounds per player per tournament to control the density of the resulting interaction network and computed the classical R-metric for each configuration. For each season, approximately 40 PGA Tour tournaments are included, and the values shown in Figure 5 represent the yearly R-values averaged across all tournaments within each season.

Figure 5 demonstrates a clear convergence behavior of the R-metric as a function of artificial pairing density. At very low densities (1–3 rounds), R is both low and highly unstable, reflecting a noise-dominated regime that would misleadingly suggest a luck-driven structure. As pairing density increases, the median R rises rapidly and the dispersion contracts. Beyond approximately 10 pairing rounds per tournament, both the median and variance of R stabilize near 0.85–0.90, indicating statistical convergence.

This result implies that the apparent randomness of golf observed in sparse head-to-head constructions is primarily a consequence of structural interaction sparsity. Once this limitation is removed through sufficiently dense Artificial Round-Robin construction, professional golf exhibits strong and stable skill dominance, positioning it within the high-skill regime of the unified skill–luck spectrum.

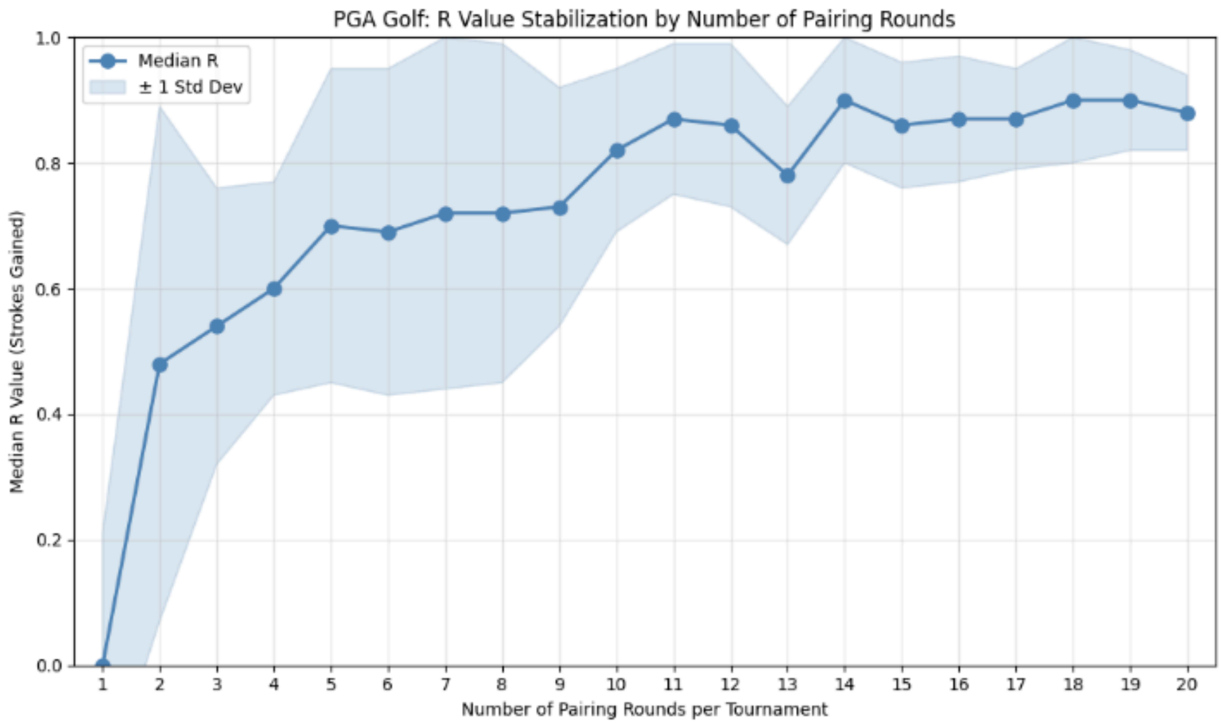

Figure 5. Stabilization of the R-metric in PGA Tour Golf as a function of artificial pairing density. As pairing rounds increase, the median R converges toward 0.9 and dispersion decreases, indicating recovery of the latent skill-dominant structure of stroke-play golf.

4. Discussion

This section interprets the empirical findings of the proposed Generalized R-metric and the Artificial Round-Robin framework from a structural and governance perspective. Building on the cross-league and cross-sport results presented in Section 3, we first identify the primary structural and sporting factors that systematically influence R-values across competitions. We then demonstrate how the Generalized R framework can be used as a quantitative decision-support tool for real-world governance through case studies on the UEFA Champions League and K League.

4.1. Primary Influence Factors on R

Previous research on the skill–luck trade-off has largely focused on outcome stability across repeated competitions, yet the observed balance is shaped by multiple structural and sport-specific factors. Building on our empirical results, we summarize the primary influence factors on the R-metric as follows:

- **Tournament Format:** Round-robin formats naturally produce higher R values than knockout tournaments because they allow more opportunities for the better team to demonstrate their superiority. The law of large numbers ensures that as the number of matches increases, random variation diminishes and true skill differences become more apparent. This principle also explains why leagues with longer seasons, such as Major

League Baseball's 162-game schedule compared to most football leagues' 30-40 match seasons, should in theory exhibit higher R values, all else equal.

- **Salary Cap Regulations:** Leagues with stricter salary and spending restrictions should exhibit lower R values due to increased competitive parity. These regulations redistribute talent more evenly across teams, narrowing the gap between the best and worst squads and making it more difficult for superior teams to consistently dominate. Salary caps also impede team continuity, as successful organizations face greater challenges retaining their best players over multiple seasons.
- **Transfer Rules:** More fluid transfer markets and fewer roster restrictions should increase R values by allowing stronger teams to acquire top talent more easily. Alternatively, regulations such as homegrown player quotas, foreign player limits, and roster size restrictions create obstacles that prevent talent from concentrating in elite teams. These barriers reduce the skill gap between teams and introduce additional uncertainty into competitive outcomes.
- **Team Fluctuations:** Teams experience performance volatility throughout a season due to factors such as injuries, managerial changes, tactical adjustments, and even weather variations that affect playing conditions. These fluctuations introduce noise into performance measurements, making it harder to isolate persistent skill differences from temporary disruptions. Lower levels of intra-season fluctuation should therefore produce higher R values, as team quality remains more stable and predictable over time.
- **Home Advantage:** Home advantage acts as a confounding factor that adds noise to performance measurements and generally decreases R values. In symmetric league formats where teams play each opponent once at home and once away, home advantage is balanced but still makes it more difficult for the better team to win any individual match. Asymmetric schedules, common in American sports leagues, compound this issue since teams play equal home and away games but against different opponents, while international tournaments at neutral sites introduce further complications through unequal fan support and travel conditions.
- **The True Influence of Skill and Luck:** Beyond structural factors, the fundamental nature of different sports determines how effectively skill translates into competitive outcomes. As Bauer and Bauer (2025) demonstrate in tennis [11], relatively small skill advantages can project into dramatically larger outcome differences, as the top players win about 53% of their points, which translates to winning 78% of their matches. This amplification mechanism varies considerably across sports, as in competitions where minor skill edges produce disproportionate outcome advantages, R values should be correspondingly higher. However, sports where skill advantages translate more linearly into results will exhibit more moderate R values, even when the underlying skill distribution across competitors is similar and all other factors are equal.

While it is difficult to isolate each factor's effect on the R value, these factors collectively help interpret the systematic variation observed across football leagues and global sports, as well as the broader business, fan engagement, and sporting trade-offs that league offices and federations must balance.

4.2. Strategic Applications

Beyond descriptive analysis, the Generalized R-metric serves as a practical decision-support tool for strategic governance. Leagues and federations often face the challenge of implementing structural reforms without quantitative benchmarks for competitive balance. Our framework addresses this gap by providing an objective basis for evaluating major structural changes.

4.2.1. League Format Evaluation (UCL Case)

A primary application of our framework is the assessment of recent structural reforms, such as the 2024/25 overhaul of the UEFA Champions League. In the revised design, the traditional group stage of 32 teams in eight groups of four has been replaced by a single 36-team league phase in which each club plays eight matches against eight different opponents (four home, four away), rather than facing the same three opponents twice.

Figure 6 reports the temporal evolution of the Generalized R-value for the UEFA Champions League over recent seasons, with the vertical demarcation indicating the transition from the traditional group-stage format (up to 2023–24) to the new league-phase structure introduced from 2024–25 onward. Under the former format, R-values remain persistently low and volatile, typically ranging between 0.20 and 0.55, reflecting strong sensitivity to draw effects and limited interaction density. Following the introduction of the new league-phase format, the Generalized R exhibits a pronounced upward shift, rising to approximately 0.60 in 2024–25 and further toward 0.80 in 2025–26.

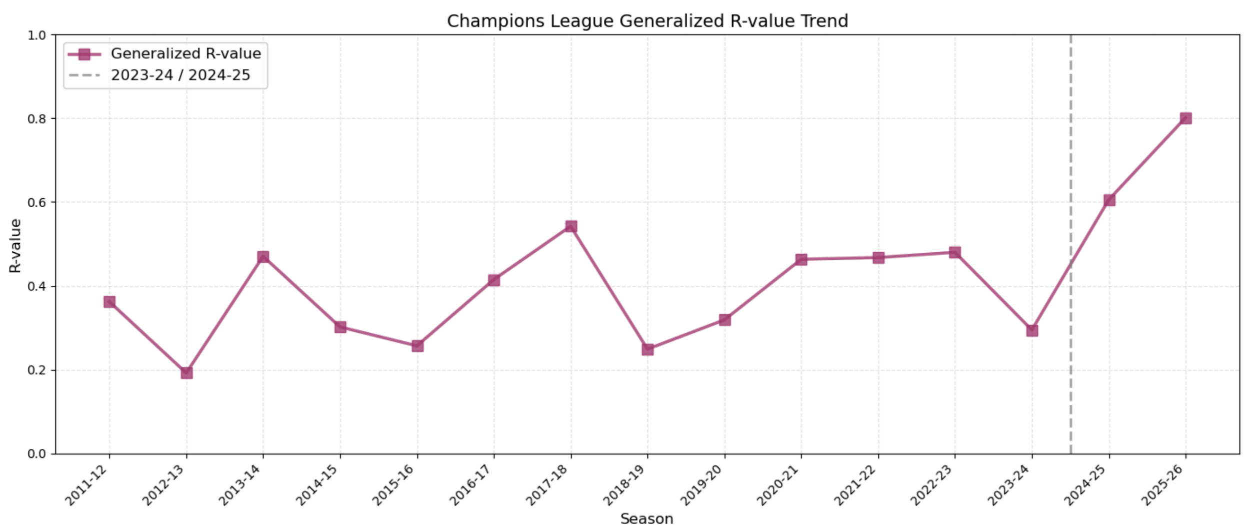

Figure 6. Temporal evolution of the Generalized R-value in the UEFA Champions League, with the vertical line marking the transition from the group-stage format to the new league-phase structure in 2024–25.

Although longer-term stabilization will require additional seasons of data, this early pattern is consistent with the hypothesis that increased opponent diversity and interaction density reduce schedule-induced variance and enhance the degree to which intrinsic team strength is reflected in outcomes. In this sense, the Generalized R-metric provides a practical ex-post diagnostic tool for quantitatively evaluating whether structural reforms achieve their intended competitive-balance objectives.

4.2.2. League Expansion Strategy (K League Case)

Figure 7 shows the recent evolution of the Generalized R-value in the K League from 2021 to 2025. Relative to major European leagues, the K League exhibits persistently lower and more volatile R-values, indicating a competitive environment in which match outcomes are more strongly influenced by randomness than by sustained team strength. The sharp drop in R observed in 2024, followed by only partial recovery in 2025, highlights the structural instability of the current competitive balance.

From a governance perspective, this has important strategic implications. A low and unstable R implies that superior performance is not consistently rewarded, which may weaken long-term investment incentives for clubs. League expansion provides a direct structural lever to address this issue by increasing match volume and interaction density. Using the Generalized R-metric, alternative league sizes can be evaluated ex ante to assess whether expansion would shift the K League toward a more skill-dominant competitive regime without excessive talent dilution.

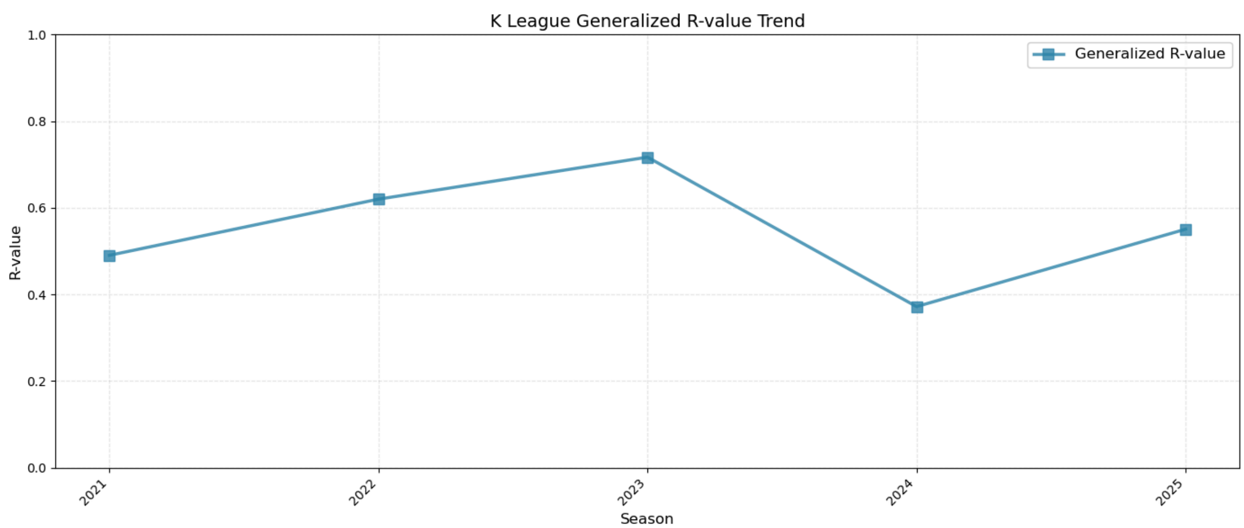

Figure 7**.** Temporal evolution of the Generalized R-value in the K League from 2021 to 2025. The series exhibits persistently low and volatile R-values relative to major European leagues, indicating a competitive environment with elevated sensitivity to random variance.

Future applications extend beyond revenue optimization. Real-time R-value monitoring could detect competitive integrity anomalies, rule changes can be evaluated through their impact on competitive balance, and roster regulations (size limits, foreign player quotas) can be calibrated toward strategic R-value targets. As competition for attention intensifies across entertainment options, and analytics continues to expand in the world of sports business, it will become more relevant to understand where a sport or league sits on the skill-luck spectrum, the quantitative factors that drive R-values, and the qualitative effects of the R-values on the business of sports.

5. Conclusion

This study introduces a Generalized R-metric that extends classical skill–luck decomposition to unbalanced leagues, tournaments, and individual sports. By combining opponent-strength–based stratified splitting with Artificial Round-Robin benchmarking, we provide a unified framework that reliably recovers latent skill under schedule imbalance and data sparsity. Validation on synthetic data, balanced and unbalanced team sports, and individual sports such as tennis and golf demonstrates the robustness and general applicability of the proposed method. The results confirm that the Generalized R enables consistent and comparable quantification of skill and randomness across diverse competitive structures in world football and many other sports.

As a direction for future work, while this study relies on a standard ELO formulation for opponent-strength estimation, we plan to conduct systematic sensitivity analyses with respect to ELO parameter settings and to incorporate official ranking systems or enhanced ELO variants to further improve the precision and stability of skill estimation.

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Appendix

A. Data Description

The dataset encompasses major European soccer leagues, the Handball-Bundesliga (HBL), and North American sports (NBA, MLB, NFL) to establish a baseline in balanced formats. Crucially, specific competitions were selected to test the framework under structural imbalance: the UEFA Champions League (UCL) serves as a validation case for knockout tournament formats, while K League 1 is included to assess the metric's efficacy in split-system (i.e., unbalanced) leagues. For individual sports, we incorporated ATP Tennis and PGA Tour Golf data. Across all team sports, the dataset contains match-level results with unique team identifiers, season information, home and away teams, and venue designation. For individual sports, tournament-level results include player identifiers, finishing positions, and total stroke or match outcomes for each event.

Sport	Competition	Format	Period	# Participants
Soccer	Premier League (ENG)	Full League	2015/16 - 2024/25	20
Soccer	EFL Championship (ENG)	Full League	2015/16 - 2024/25	24
Soccer	La Liga (ESP)	Full League	2015/16 - 2024/25	20
Soccer	La Liga 2 (ESP)	Full League	2015/16 - 2024/25	22
Soccer	Bundesliga 1 (GER)	Full League	2015/16 - 2024/25	18
Soccer	Bundesliga 2 (GER)	Full League	2015/16 - 2024/25	18
Soccer	Serie A (ITA)	Full League	2015/16 - 2024/25	20
Soccer	Serie B (ITA)	Full League	2015/16 - 2024/25	20
Soccer	MLS (USA)	Full League	2015/16 - 2024/25	30
Soccer	UEFA	Group +	2015/16 -	32/36

	Champions League	Knockout	2024/25	
Soccer	K League 1 (KOR)	Split System	2021 - 2025	12
Baseball	MLB (USA)	Full League	2011 - 2020	30
Basketball	NBA (USA)	Full League	2013 - 2022	30
American Football	NFL (USA)	Full League	2017 - 2025	32
Handball	HBL (Germany)	Full League	2014/15 - 2023/24	18
Tennis	ATP Tour	Head-to-Head (Individual)	2014- 2023	40 (Selected)
Golf	PGA Tour	Stroke Play (Individual)	2015 - 2022	20 (Selected)

Table A1. Summary of the comprehensive dataset.

B. Synthetic Simulation Setup for Schedule Imbalance Experiments

We simulate a synthetic league consisting of $N = 20$ agents. Each agent i is assigned a fixed latent intrinsic strength parameter $w_i \in (0, 1)$, evenly spaced in the interval $[0.05, 0.95]$. These parameters are used solely to define the ground-truth pairwise win probabilities and are not estimated during the experiment. Since all match outcomes are generated from stationary, time-invariant win probabilities, the true skill–luck ratio satisfies $R_{true} \approx 1.0$ by construction.

Pairwise match outcomes are generated using a Bradley–Terry–type formulation. For any two teams i and j with intrinsic strengths w_i and w_j, the win probability of team i is given by

$$p(i > j) = \frac{w_i(1-w_j)}{w_i(1-w_j) + w_j(1-w_i)}$$

Match outcomes are sampled independently from the corresponding Bernoulli distribution. Elo ratings are not used to generate outcomes and serve no role in the ground-truth construction.

Opponent strengths are discretized into $L = 19$ evenly spaced levels on the interval $[0.1, 0.9]$. For each match, opponents are sampled from this grid with replacement. Each simulated season consists of $T = 38$ matches per team.

To control structural bias in the schedule, we introduce a Degree of Schedule Imbalance parameter $\delta \in [0, 1]$. For $\delta = 0$, opponents are assigned uniformly at random across the season (balanced schedule). For $\delta = 1$, , a maximally imbalanced "heaven–hell" schedule is constructed, where teams face weak opponents in the first half and strong opponents in the second half. For intermediate values $0 < \delta < 1$, the first half of the season is increasingly populated by weaker opponents while the second half is increasingly populated by stronger opponents. Specifically, as δ increases, the pool of opponents for the first half is progressively restricted to the lower end of the opponent-strength distribution, while the pool for the second half is restricted to the upper end. This creates a smooth transition from a fully random schedule ($\delta = 0$) to a fully sorted "heaven–hell" schedule ($\delta = 1$).

For each (w_i, δ) configuration, matches are split using two procedures: (i) Getty's temporal split, which partitions matches chronologically into first and second halves of the season, and (ii) Generalized opponent-strength–based stratified splitting, which sorts matches by opponent difficulty and alternately assigns them to two groups so that both contain a balanced mixture of easy and difficult opponents. Mean performances P and Q are computed for each split and used to calculate the corresponding R-values.

Each (w_i, δ) configuration is repeated over 10 independent Monte Carlo runs with distinct random seeds. Reported values correspond to the mean and standard deviation across repetitions.

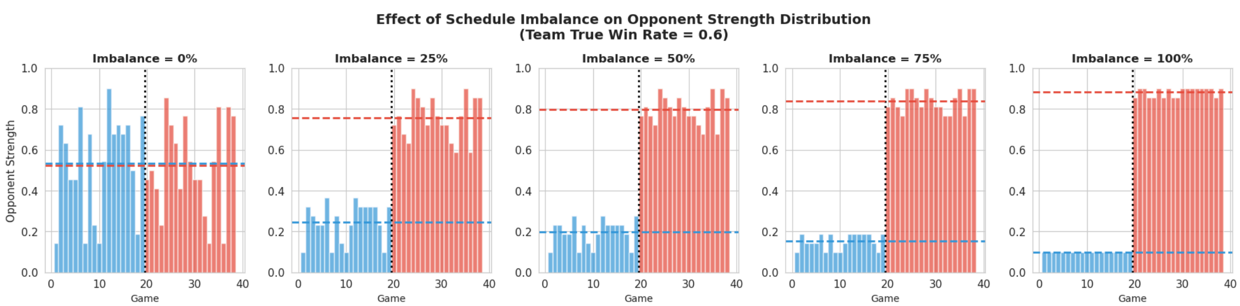

Figure B1. Effect of the schedule imbalance parameter δ on the distribution of opponent strength over a season. As δ increases from 0 to 1, opponents in the first half are progressively sampled from weaker teams, while opponents in the second half are sampled from stronger teams, yielding a smooth transition from a fully random schedule ($\delta = 0$) to a fully sorted "heaven–hell" schedule ($\delta = 1$).

Figure B1. visualizes the opponent strength distribution over a season for increasing values of δ for a representative team with $w_{true} = 0.6$. As δincreases, opponent difficulty becomes progressively sorted from weak to strong across the season, inducing artificial temporal performance trends despite time-invariant intrinsic team skill. Under an unbiased splitting procedure, the expected performance difference between splits is zero, implying $R_{true} = 1.0$; any deviation from unity therefore reflects methodological bias induced solely by schedule construction and splitting.